Disaggregated: OLS with HAC standard errors

Specification	Disaggregated	Number of obs	2,671
Dependent variable	Level change in e (USD per NZD)	Wald chi2(27)	70.28
Sample	2014-01-09 to 2025-01-31	Prob > chi2	0.0000
Std. errors	HAC (Newey–West), 10 lags	R-squared	0.0130
Condition no.	203.3	Adj R-squared	0.0029
AIC / BIC	-21196.9 / -21032.0	Root MSE	0.004552

	Coef.	HAC Std. Err.	z	$P>\|z\|$	[95% Conf.	Interval]
US Issuance (k=-5, post)	2.575×10^{-6}	1.722×10^{-6}	1.50	0.135	-8.003×10^{-7}	5.950×10^{-6}
US Issuance (k=-4, post)	-2.839×10^{-6}	1.864×10^{-6}	−1.52	0.128	-6.493×10^{-6}	8.144×10^{-7}
US Issuance (k=-3, post)	-1.709×10^{-6}	1.544×10^{-6}	−1.11	0.268	-4.735×10^{-6}	1.318×10^{-6}
US Issuance (k=-2, post)	1.599×10^{-6}	1.663×10^{-6}	0.96	0.336	-1.661×10^{-6}	4.859×10^{-6}
US Issuance (k=-1, post)	-1.931×10^{-6}	2.262×10^{-6}	−0.85	0.393	-6.365×10^{-6}	2.503×10^{-6}
US Issuance (k=+0, event)	1.549×10^{-6}	1.916×10^{-6}	0.81	0.419	-2.206×10^{-6}	5.305×10^{-6}
US Issuance (k=+1, pre)	5.219×10^{-6}**	2.267×10^{-6}	2.30	0.021	7.765×10^{-7}	9.662×10^{-6}
US Issuance (k=+2, pre)	3.796×10^{-6}**	1.795×10^{-6}	2.11	0.034	2.775×10^{-7}	7.314×10^{-6}
US Issuance (k=+3, pre)	-3.565×10^{-7}	1.402×10^{-6}	−0.25	0.799	-3.104×10^{-6}	2.391×10^{-6}
US Issuance (k=+4, pre)	2.768×10^{-6}*	1.608×10^{-6}	1.72	0.085	-3.833×10^{-7}	5.920×10^{-6}
US Issuance (k=+5, pre)	-1.963×10^{-7}	1.695×10^{-6}	−0.12	0.908	-3.519×10^{-6}	3.126×10^{-6}
US Principal (k=-5, post)	6.203×10^{-7}	1.032×10^{-6}	0.60	0.548	-1.403×10^{-6}	2.644×10^{-6}
US Principal (k=-4, post)	-1.321×10^{-6}	2.125×10^{-6}	−0.62	0.534	-5.486×10^{-6}	2.844×10^{-6}
US Principal (k=-3, post)	-3.885×10^{-7}	1.061×10^{-6}	−0.37	0.714	-2.468×10^{-6}	1.691×10^{-6}
US Principal (k=-2, post)	7.371×10^{-7}	1.538×10^{-6}	0.48	0.632	-2.277×10^{-6}	3.752×10^{-6}
US Principal (k=-1, post)	5.455×10^{-6}**	2.143×10^{-6}	2.55	0.011	1.255×10^{-6}	9.656×10^{-6}
US Principal (k=+0, event)	-1.567×10^{-6}	2.035×10^{-6}	−0.77	0.441	-5.555×10^{-6}	2.422×10^{-6}
US Principal (k=+1, pre)	9.502×10^{-7}	3.509×10^{-6}	0.27	0.787	-5.928×10^{-6}	7.828×10^{-6}
US Principal (k=+2, pre)	8.383×10^{-7}	3.473×10^{-6}	0.24	0.809	-5.968×10^{-6}	7.645×10^{-6}
US Principal (k=+3, pre)	-1.960×10^{-6}	2.273×10^{-6}	−0.86	0.388	-6.414×10^{-6}	2.494×10^{-6}
US Principal (k=+4, pre)	6.644×10^{-7}	1.491×10^{-6}	0.45	0.656	-2.257×10^{-6}	3.586×10^{-6}
US Principal (k=+5, pre)	3.875×10^{-6}**	1.623×10^{-6}	2.39	0.017	6.936×10^{-7}	7.057×10^{-6}
US coupon	9.588×10^{-6}	1.053×10^{-5}	0.91	0.362	-1.104×10^{-5}	3.022×10^{-5}
US buyback	1.903×10^{-6}	7.607×10^{-6}	0.25	0.802	-1.301×10^{-5}	1.681×10^{-5}
VIX	-3.550×10^{-5}*	1.922×10^{-5}	−1.85	0.065	-7.317×10^{-5}	2.166×10^{-6}
Treasury swap	-5.214×10^{-5}	7.122×10^{-5}	−0.73	0.464	-1.917×10^{-4}	8.745×10^{-5}
US release	3.328×10^{-4}	3.376×10^{-4}	0.99	0.324	-3.288×10^{-4}	9.944×10^{-4}
Constant	4.178×10^{-4}	3.225×10^{-4}	1.30	0.195	-2.142×10^{-4}	1.050×10^{-3}

* $p<0.10$, ** $p<0.05$, *** $p<0.01$ (two-sided where applicable).

Residual diagnostics

Test	Statistic	p-value	Null hypothesis
Breusch–Pagan LM (heteroskedasticity)	176.385	<0.001	homoskedastic residuals
White LM (heteroskedasticity)	338.410	<0.001	homoskedastic residuals
Breusch–Godfrey LM (10 lags; autocorrelation)	21.257	0.019	no serial correlation
Ljung–Box Q (10 lags; autocorrelation)	21.377	0.019	no serial correlation
Jarque–Bera (normality)	230.127	<0.001	normal residuals
Durbin–Watson	2.092		DW ≈ 2: no first-order autocorrelation

Specification	Disaggregated	Number of obs	2,671
Dependent variable	100 x log change in e (USD per NZD)	Wald chi2(27)	75.27
Sample	2014-01-09 to 2025-01-31	Prob > chi2	0.0000
Std. errors	HAC (Newey–West), 10 lags	R-squared	0.0137
Condition no.	203.3	Adj R-squared	0.0036
AIC / BIC	5503.6 / 5668.5	Root MSE	0.6744

	Coef.	HAC Std. Err.	z	$P > \|z\|$	[95% Conf.	Interval]
US Issuance (k=-5, post)	3.955×10^{-4}	2.622×10^{-4}	1.51	0.132	-1.185×10^{-4}	9.095×10^{-4}
US Issuance (k=-4, post)	-4.240×10^{-4}	2.823×10^{-4}	−1.50	0.133	-9.772×10^{-4}	1.293×10^{-4}
US Issuance (k=-3, post)	-2.745×10^{-4}	2.406×10^{-4}	−1.14	0.254	-7.460×10^{-4}	1.970×10^{-4}
US Issuance (k=-2, post)	2.756×10^{-4}	2.595×10^{-4}	1.06	0.288	-2.330×10^{-4}	7.841×10^{-4}
US Issuance (k=-1, post)	-3.329×10^{-4}	3.471×10^{-4}	−0.96	0.338	-1.013×10^{-3}	3.475×10^{-4}
US Issuance (k=+0, event)	2.550×10^{-4}	2.986×10^{-4}	0.85	0.393	-3.303×10^{-4}	8.402×10^{-4}
US Issuance (k=+1, pre)	7.934×10^{-4}**	3.529×10^{-4}	2.25	0.025	1.017×10^{-4}	1.485×10^{-3}
US Issuance (k=+2, pre)	5.936×10^{-4}**	2.791×10^{-4}	2.13	0.033	4.660×10^{-5}	1.141×10^{-3}
US Issuance (k=+3, pre)	-7.434×10^{-5}	2.164×10^{-4}	−0.34	0.731	-4.984×10^{-4}	3.497×10^{-4}
US Issuance (k=+4, pre)	4.313×10^{-4}*	2.479×10^{-4}	1.74	0.082	-5.452×10^{-5}	9.172×10^{-4}
US Issuance (k=+5, pre)	-2.901×10^{-5}	2.648×10^{-4}	−0.11	0.913	-5.480×10^{-4}	4.900×10^{-4}
US Principal (k=-5, post)	9.296×10^{-5}	1.456×10^{-4}	0.64	0.523	-1.923×10^{-4}	3.782×10^{-4}
US Principal (k=-4, post)	-2.469×10^{-4}	3.257×10^{-4}	−0.76	0.448	-8.854×10^{-4}	3.915×10^{-4}
US Principal (k=-3, post)	-5.530×10^{-5}	1.617×10^{-4}	−0.34	0.732	-3.722×10^{-4}	2.616×10^{-4}
US Principal (k=-2, post)	1.284×10^{-4}	2.288×10^{-4}	0.56	0.575	-3.201×10^{-4}	5.769×10^{-4}
US Principal (k=-1, post)	7.504×10^{-4}***	2.768×10^{-4}	2.71	0.007	2.079×10^{-4}	1.293×10^{-3}
US Principal (k=+0, event)	-2.309×10^{-4}	3.225×10^{-4}	−0.72	0.474	-8.631×10^{-4}	4.012×10^{-4}
US Principal (k=+1, pre)	7.014×10^{-5}	5.168×10^{-4}	0.14	0.892	-9.427×10^{-4}	1.083×10^{-3}
US Principal (k=+2, pre)	1.778×10^{-4}	5.088×10^{-4}	0.35	0.727	-8.194×10^{-4}	1.175×10^{-3}
US Principal (k=+3, pre)	-2.565×10^{-4}	3.186×10^{-4}	−0.81	0.421	-8.809×10^{-4}	3.680×10^{-4}
US Principal (k=+4, pre)	1.251×10^{-4}	2.209×10^{-4}	0.57	0.571	-3.078×10^{-4}	5.580×10^{-4}
US Principal (k=+5, pre)	5.357×10^{-4}**	2.229×10^{-4}	2.40	0.016	9.884×10^{-5}	9.726×10^{-4}
US coupon	1.357×10^{-3}	1.558×10^{-3}	0.87	0.384	-1.697×10^{-3}	4.411×10^{-3}
US buyback	2.260×10^{-4}	1.042×10^{-3}	0.22	0.828	-1.817×10^{-3}	2.269×10^{-3}
VIX	-5.606×10^{-3}*	3.141×10^{-3}	−1.79	0.074	-1.176×10^{-2}	5.494×10^{-4}
Treasury swap	-5.956×10^{-3}	9.701×10^{-3}	−0.61	0.539	-2.497×10^{-2}	1.306×10^{-2}
US release	5.423×10^{-2}	5.122×10^{-2}	1.06	0.290	-4.617×10^{-2}	1.546×10^{-1}
Constant	6.636×10^{-2}	5.175×10^{-2}	1.28	0.200	-3.508×10^{-2}	1.678×10^{-1}

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided where applicable).

Residual diagnostics

Test	Statistic	p-value	Null hypothesis
Breusch–Pagan LM (heteroskedasticity)	213.271	<0.001	homoskedastic residuals
White LM (heteroskedasticity)	413.331	<0.001	homoskedastic residuals
Breusch–Godfrey LM (10 lags; autocorrelation)	25.920	0.004	no serial correlation
Ljung–Box Q (10 lags; autocorrelation)	26.100	0.004	no serial correlation
Jarque–Bera (normality)	243.077	<0.001	normal residuals
Durbin–Watson	2.099		DW ≈ 2: no first-order autocorrelation